Algebra II with Trigonometry Chapter 6.5

Study guide

Saved tutoring response with MathJax rendering. Original PDF file: 4533652_alg-2trig-h-chp-6-5-note.pdf

Certainly! Here's a thorough, self-contained study guide based on the PDF you provided.

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1. Short Summary

This document covers **inverse trigonometric functions**—sine, cosine, tangent, secant, cosecant, and cotangent. For each, it introduces the definition, domain, range, notation, cancellation properties, and their key graphs. Several worked examples and problem types, both exact and calculator-based, are included to reinforce understanding.

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2. Core Definitions, Theorems, and Formulas

Definitions and Notation

- **Inverse Sine (Arcsine):**

$$y = \sin^{-1} x \text{ or } y = \arcsin x$$

- **Domain:** $[-1, 1]$
- **Range:** $[-\frac{\pi}{2}, \frac{\pi}{2}]$
- If $y = \sin^{-1} x$, then $\sin y = x$ and $y \in [-\frac{\pi}{2}, \frac{\pi}{2}]$

- **Inverse Cosine (Arccosine):**

$$y = \cos^{-1} x \text{ or } y = \arccos x$$

- **Domain:** $[-1, 1]$

- **Range:** $[0, \pi]$
- If $y = \cos^{-1} x$, then $\cos y = x$ and $y \in [0, \pi]$

- **Inverse Tangent (Arctangent):**

$y = \tan^{-1} x$ or $y = \arctan x$

- **Domain:** $\mathbb{R}$
- **Range:** $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$
- If $y = \tan^{-1} x$, then $\tan y = x$ and $y \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$

Cancellation Properties

- **Sine:**
- $\sin(\sin^{-1} x) = x$, for $-1 \leq x \leq 1$
- $\sin^{-1}(\sin x) = x$, for $-\frac{\pi}{2} \leq x \leq \frac{\pi}{2}$
- **Cosine:**
- $\cos(\cos^{-1} x) = x$, for $-1 \leq x \leq 1$
- $\cos^{-1}(\cos x) = x$, for $0 \leq x \leq \pi$
- **Tangent:**
- $\tan(\tan^{-1} x) = x$, for $x \in \mathbb{R}$
- $\tan^{-1}(\tan x) = x$, for $-\frac{\pi}{2} < x < \frac{\pi}{2}$

Additional Inverse Functions

- **Secant, Cosecant, Cotangent:**

Definitions require restricting the domain of the original functions to make them one-to-one.

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3. Step-by-Step Study Guide

A. What is an Inverse Trig Function?

Given a trig value (like sin θ = 1/2), an **inverse trig function** helps you find the angle θ. For example, $\sin^{-1}\left(\frac{1}{2}\right)$ asks: "For what angle is sine equal to 1/2?" Answer: $\frac{\pi}{6}$.

B. The Big Three

1. Inverse Sine $(\sin^{-1} x)$

- **Input (x):** Must be between -1 and 1.
- **Output (y):** An angle in radians, between $-\frac{\pi}{2}$ and $\frac{\pi}{2}$ (quadrants I and IV).
- **Reason for range:** Sine repeats, so we pick this interval for a unique answer.

2. Inverse Cosine $(\cos^{-1} x)$

- **Input (x):** Must be between -1 and 1.
- **Output (y):** An angle between 0 and π (quadrants I and II).
- **Unique answer:** Always produces the "principal angle" for cosine.

3. Inverse Tangent $(\tan^{-1} x)$

- **Input (x):** Any real number.
- **Output (y):** An angle between $-\frac{\pi}{2}$ and $\frac{\pi}{2}$ (not including endpoints).
- **Behavior:** Works for all real inputs, returns principal value.

C. Cancellation Properties Explained

If you apply the trig function and then its inverse, or vice versa, you get back to where you started—*under certain conditions*.

- $\sin^{-1}(\sin x) = x$ only if x is within $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$
- $\cos^{-1}(\cos x) = x$ only if x is within $[0, \pi]$
- $\tan^{-1}(\tan x) = x$ only if x is within $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$

D. Graphs

- **Graphs of the inverse functions** look like reflected versions of the original trig functions (reflected over y = x), but are only defined on their principal intervals.

E. Calculator Practice

- Use your calculator’s inverse trig buttons to find decimal approximations for real values (be sure it’s in radian mode unless otherwise specified).

F. The "Other Three"

- **Secant, cosecant, cotangent:** Inverses also exist, but their defining intervals and domains are specially chosen so the function is one-to-one.

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4. Practice Problems With Answers

Problem 1

Find the exact value: $\sin^{-1}\left(-\frac{1}{2}\right)$

Answer: $-\frac{\pi}{6}$

Problem 2

Find the exact value: $\cos^{-1}(0)$

Answer: $\frac{\pi}{2}$

Problem 3

If $\tan^{-1}(1) = y$, what is y?

Answer: $\frac{\pi}{4}$

Problem 4

Evaluate $\cos\left(\sin^{-1}(x)\right)$ as an algebraic expression in terms of x.

Answer: $\sqrt{1 - x^2}$ for $-1 \leq x \leq 1$

Problem 5

Find an approximate value: $\sin^{-1}(0.75)$ (to 5 decimal places).

Answer: 0.84806 (in radians)

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5. Assumptions or Ambiguities

- The notation $\sin^{-1}(x)$, $\cos^{-1}(x)$, and $\tan^{-1}(x)$ refers to the inverse functions (not reciprocals).
- Exact values are assumed to be in radians, the standard in higher mathematics.
- Calculators must be in radian mode to match the context unless degrees are specified.
- For algebraic conversions like $\cos(\sin^{-1} x)$, knowledge of right triangle relationships is required: if $\sin\theta = x$, then $\cos\theta = \sqrt{1-x^2}$ for non-negative cosine.
- Definitions of inverse secant, cosecant, cotangent were given only in outline, without explicit formulas; no practice problems rely on these as a result.

End of Study Guide

Let me know if you'd like further example solutions or deeper explanations on any section!